Document Understanding with Open Multimodal Models

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TNG Dev Day | Munich, 2026

Agenda

- Intro to LLM-based agents
- Intro to Multimodality
- Intuitions about embeddings and decoders
- Overview of Nemotron 3 Nano Omni
- Providing context with Nemotron Parse 2.0
- Experimenting with the Notebooks



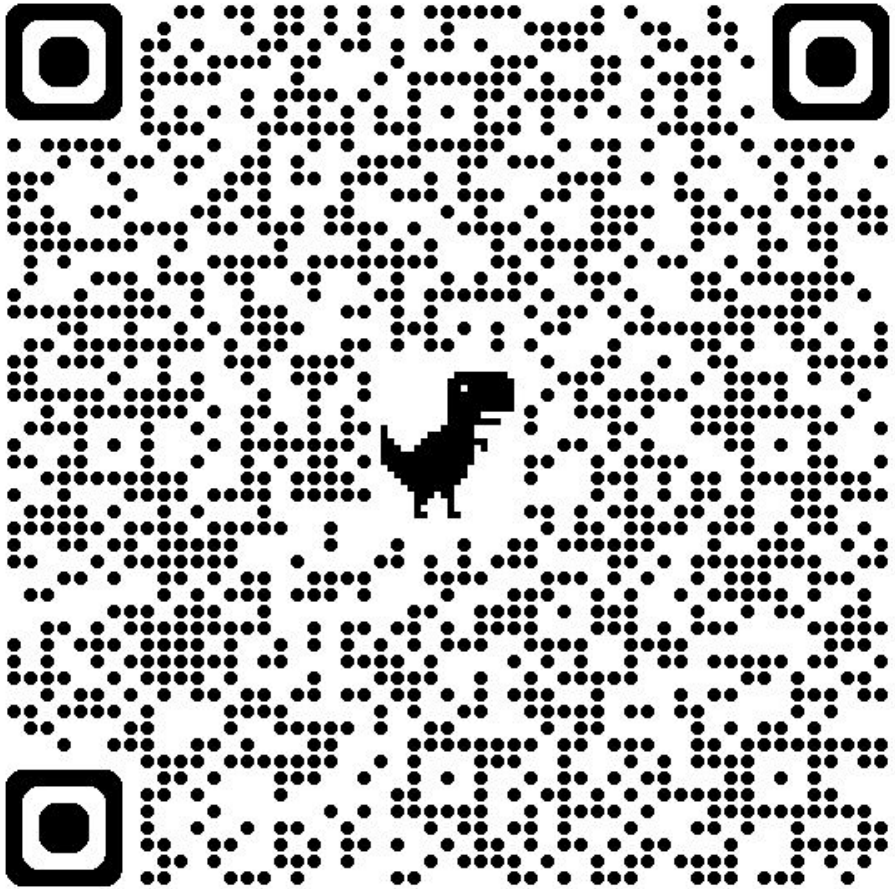
Notebooks

- Using Nemotron models through Nvidia Inference Microservices
- Using Nemotron 3 Omni + Nemotron Parse 1.2 for document understanding
- Parsing experiments with old Britannica
- A Nemotron + LangGraph agent for word counting

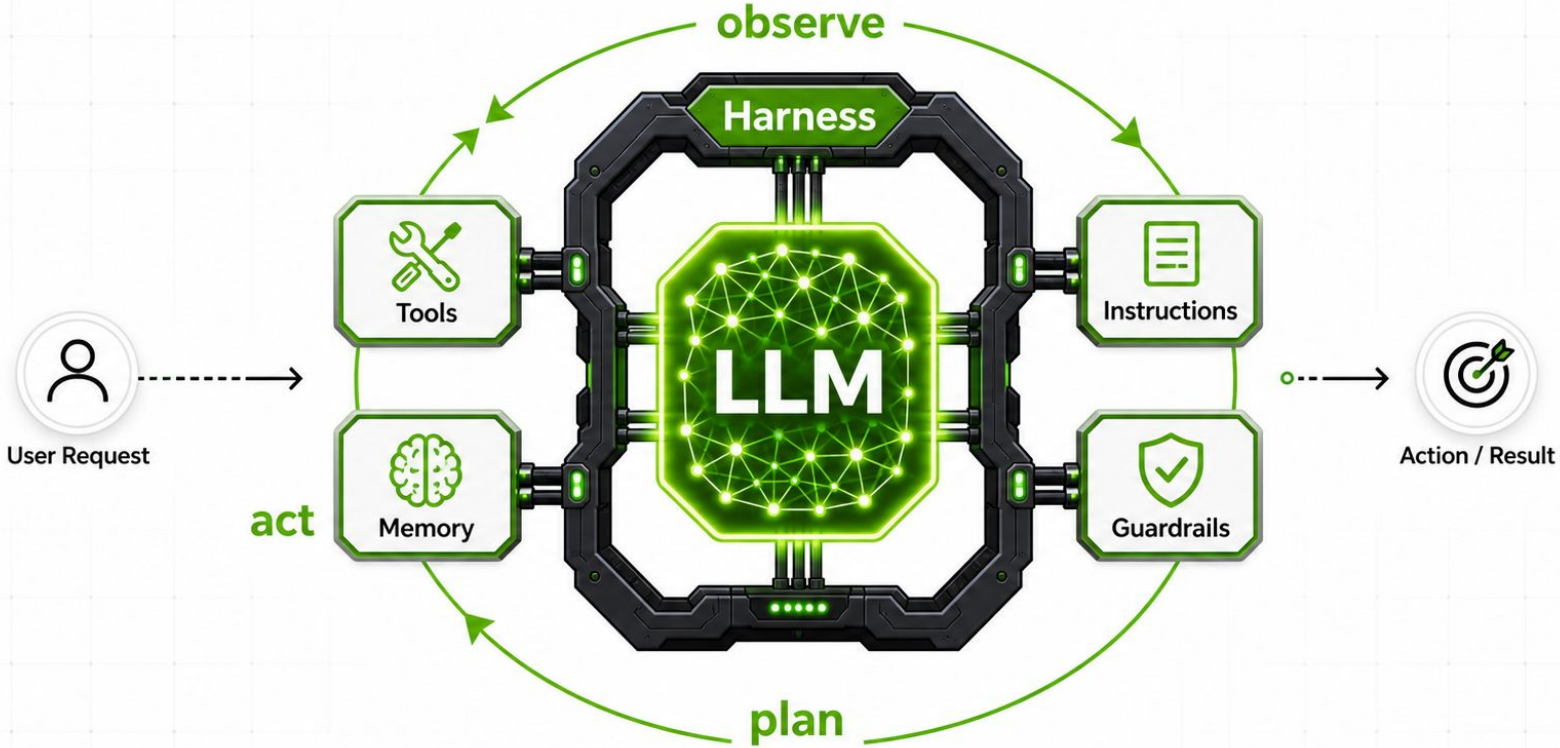
Setup

- Configuring your NVIDIA_API_KEY to use models from build.nvidia.com

Slides



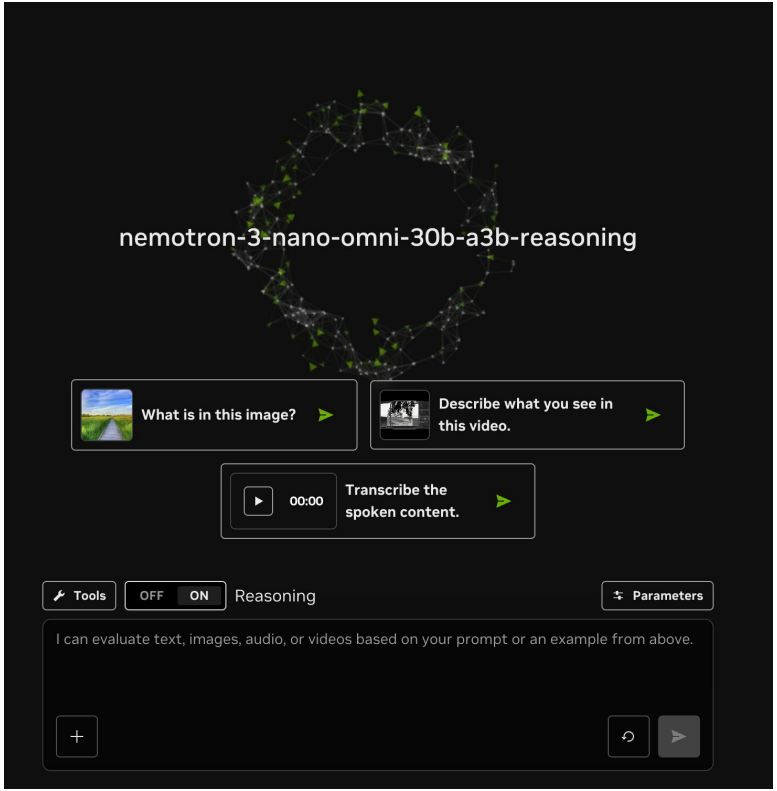
AI Agent = LLM + Harness



ReAct: Synergizing Reasoning and Acting in Language Models

Nemotron 3 Nano Omni

Reasoning VLM trained on tool calling



nemotron-3-nano-omni-30b-a3b-reasoning (live demo)

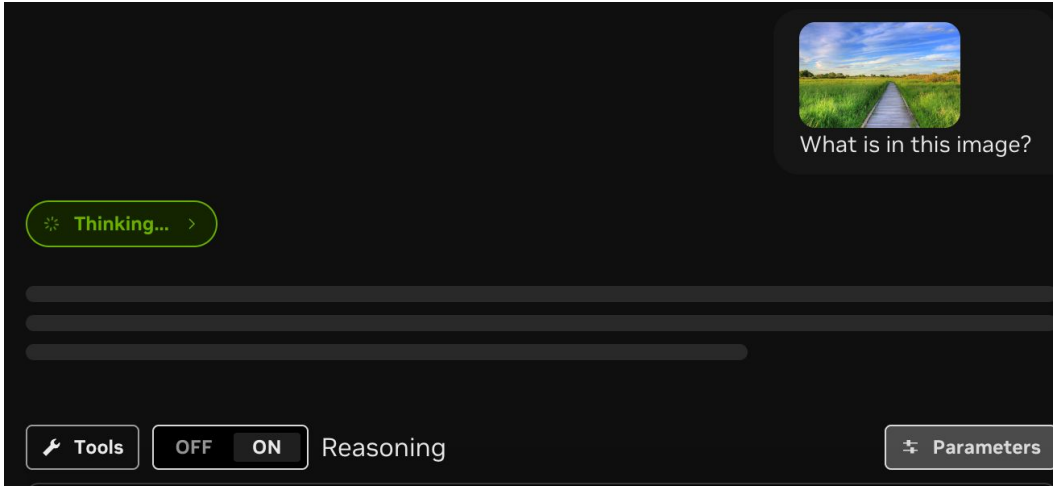
Nemotron 3 Nano Omni: Efficient and Open Multimodal Intelligence (ArXiv)

Describing Multimodal Data with Nemotron 3 Nano Omni

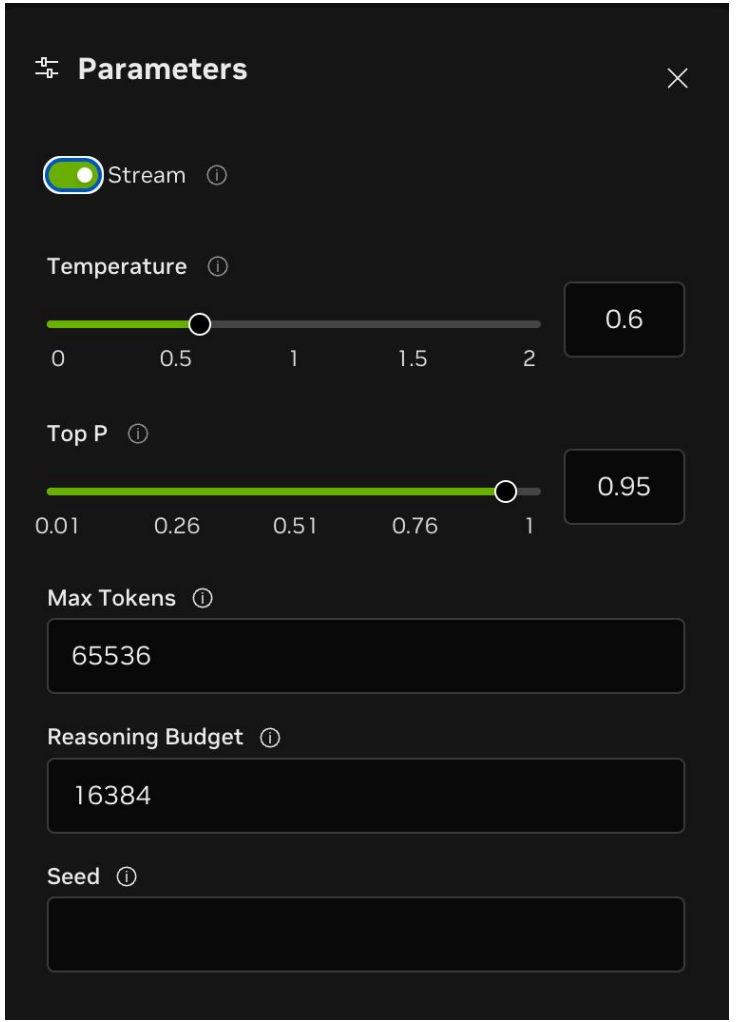


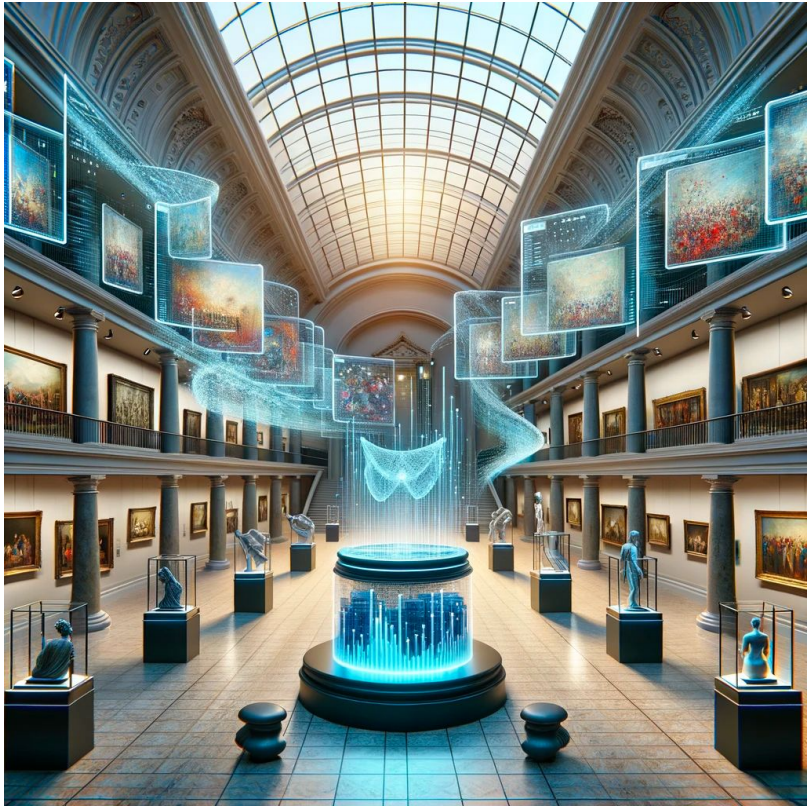
"A galloping horse, seen from the side, with the jockey riding it"

Understand the architecture to understand its parameters



https://build.nvidia.com/nvidia/nemotron-3-nano-omni-30b-a3b-reasoning



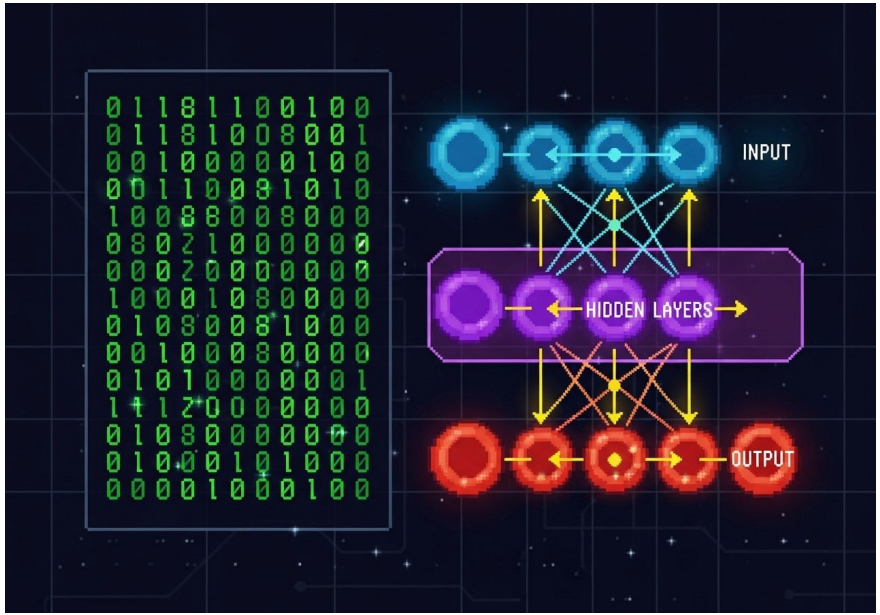


Use cases

- Video search and summarization 📽️
- Multimodal document understanding 📚
- Audio transcription 🗣️
- Dataset curation 🛠️

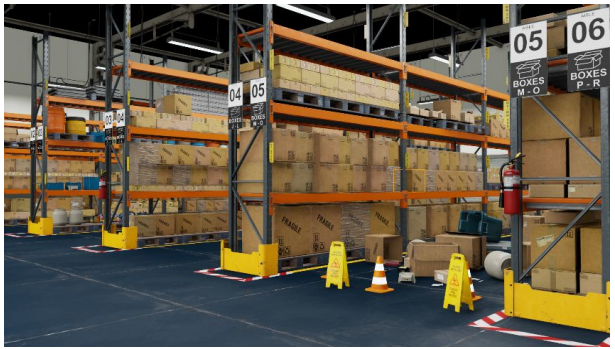
Working Hypothesis

Any data source that can be turned into a matrix can be used to train and run inference on a neural network

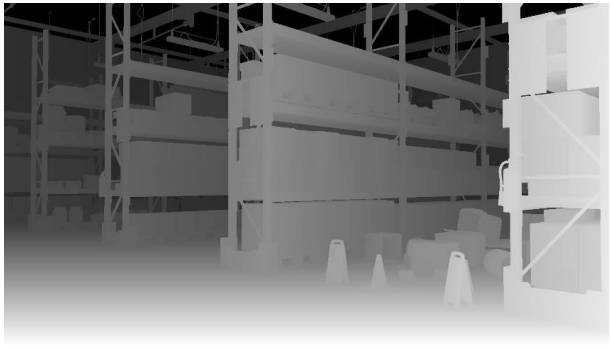


Sight

How A Computer Might See



RGB



Depth
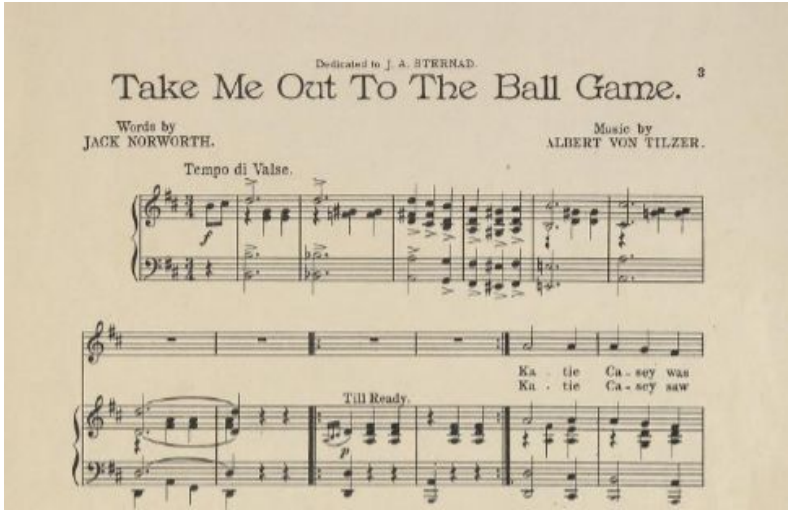


Fisheye

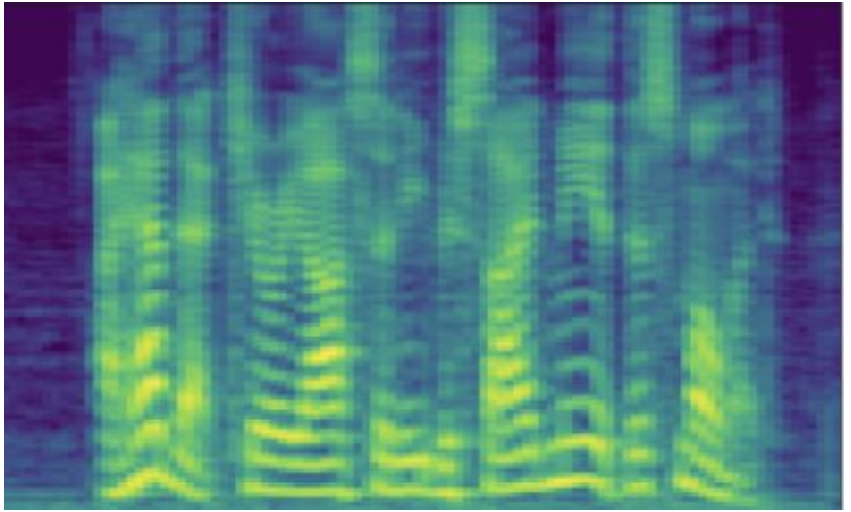
Hearing

How A Computer Might Hear

Sheet Music



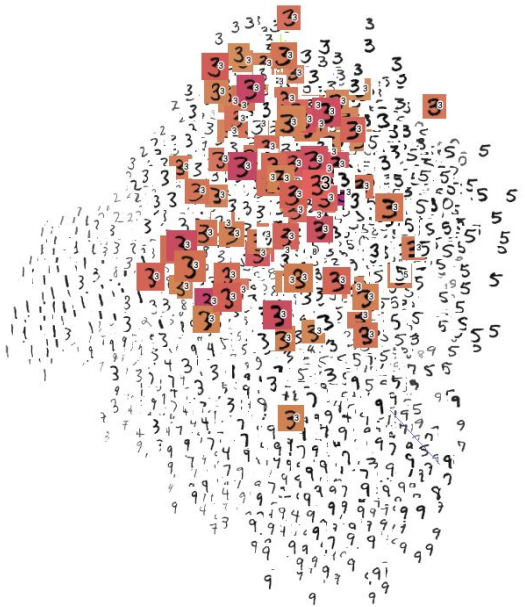
Spectrogram

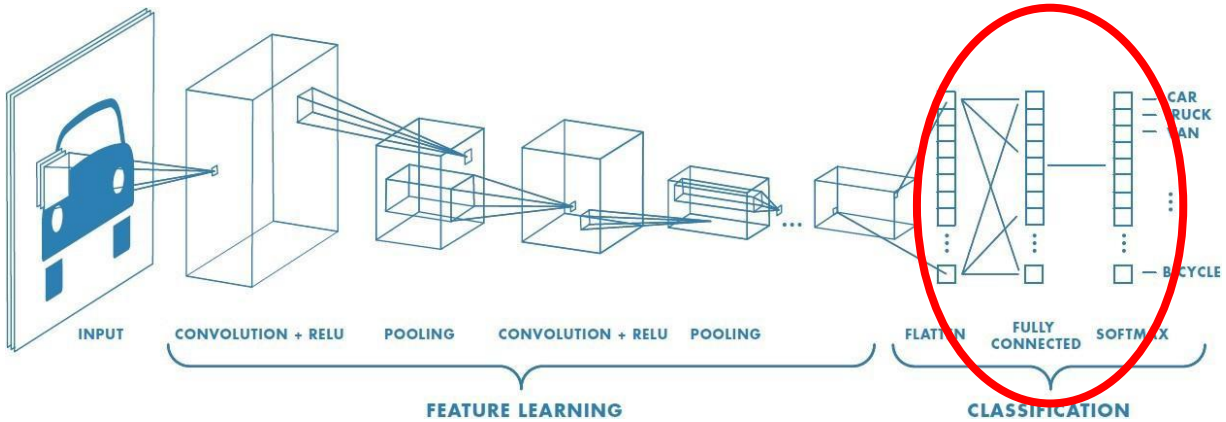


NVIDIA.

Embeddings as the enabler of multimodality

When we train a neural network, we make it **embed** inputs that are semantically similar, close to each other

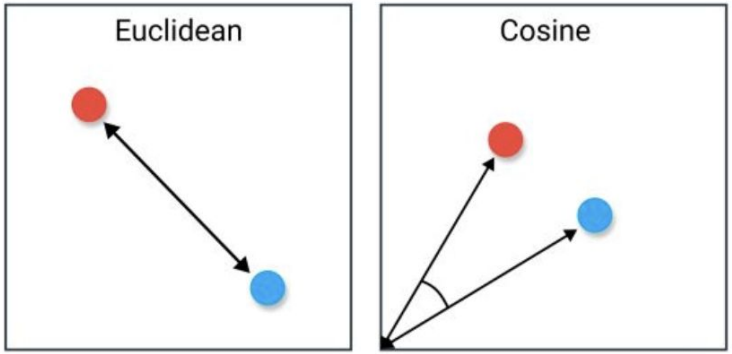




Dense vector embeddings can be produced by flattening, pooling, or feeding (learned) features to linear layers

Source of image (try exploring interactively)

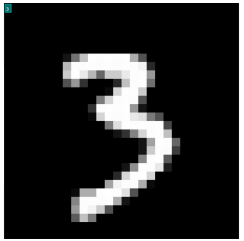
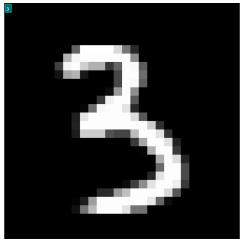
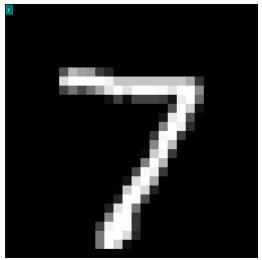
Common metrics to compare embeddings



$$d_{\text{euclid}}(x_1, x_2) = \sqrt{\sum_{i=1}^{n}(x_{1i} - x_{2i})^2}$$

$$\text{cosine similarity}(x_1, x_2) = \frac{x_1 \cdot x_2}{\|x_1\| \, \|x_2\|}$$

Similar objects have similar embeddings

$\mathbf{x}_1 =$  $\mathbf{x}_2 =$  $\mathbf{x}_3 =$ 

Similar digits (X_1, X_2): High cos similarity (≈ 0.98), Low Euclidean distance (≈ 0.3)

Different digits (X_1, X_3): Low cos similarity (≈ 0.4), High Euclidean distance (≈ 2.1)

README.md

Art Recommendation system

This is the repository of a portfolio project at DSR. This project aims to identify similar images using pre-trained computer vision networks. For an explanation of the technology see the technology section.

Contributors

- Catarina Ferreira
- Gargi Maheshwari



WIKIART
VISUAL ART ENCYCLOPEDIA

https://github.com/gargimaheshwari/Wikiart-similar-art















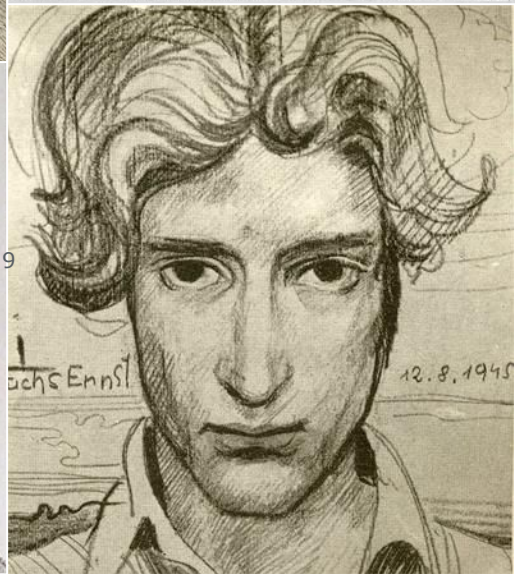




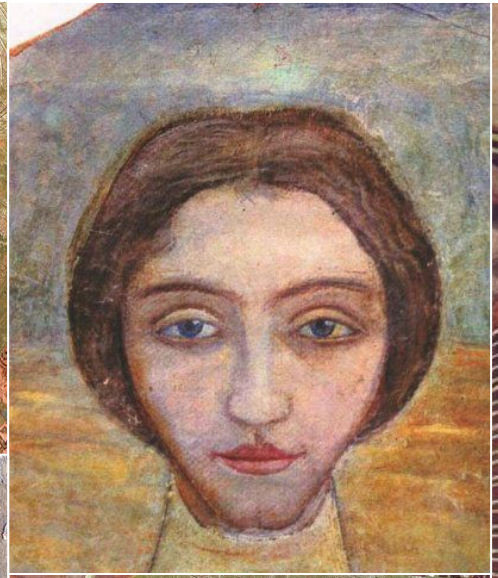




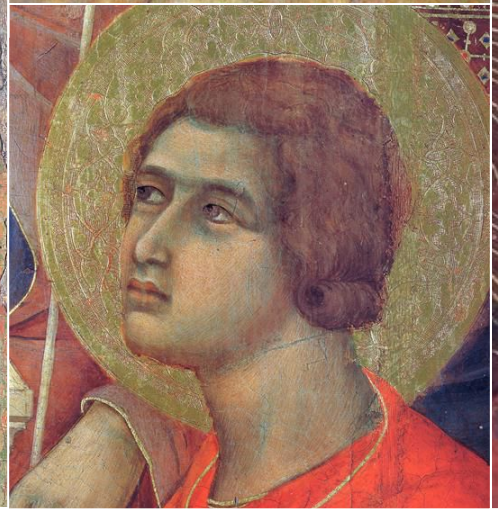


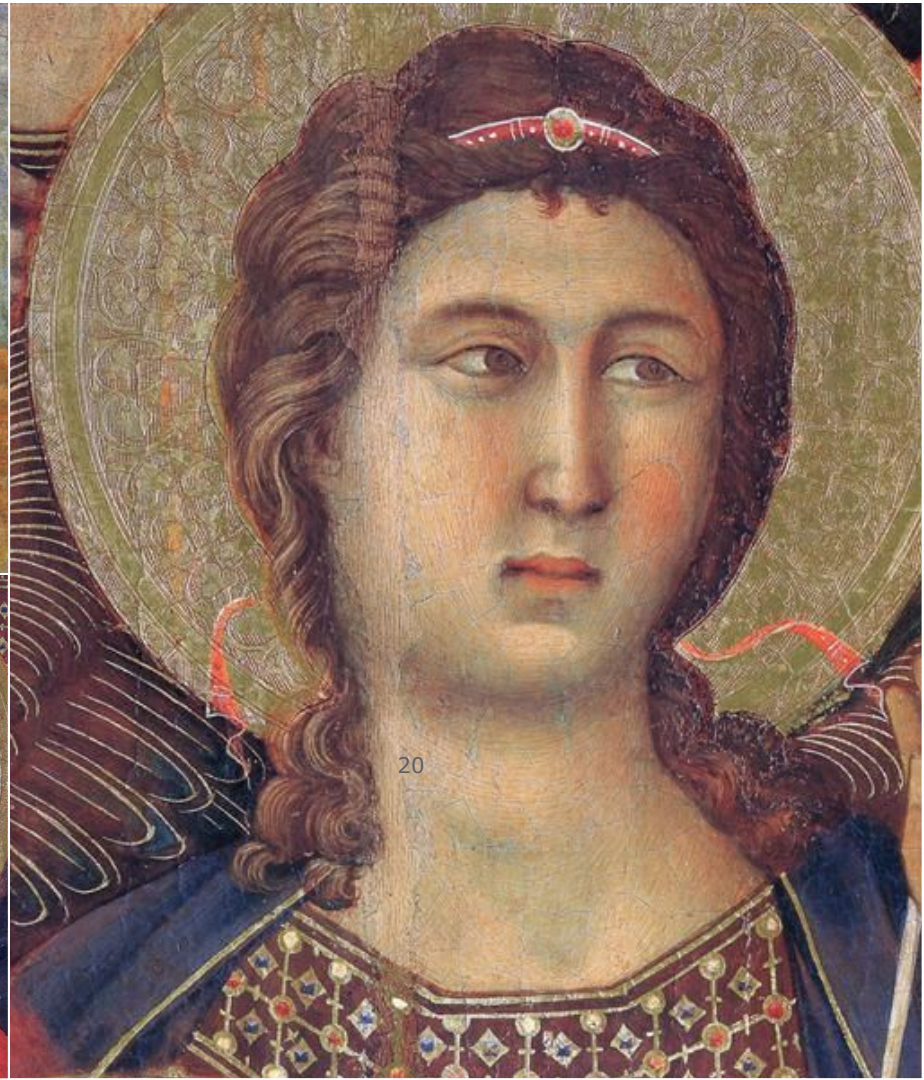

12.8.1945



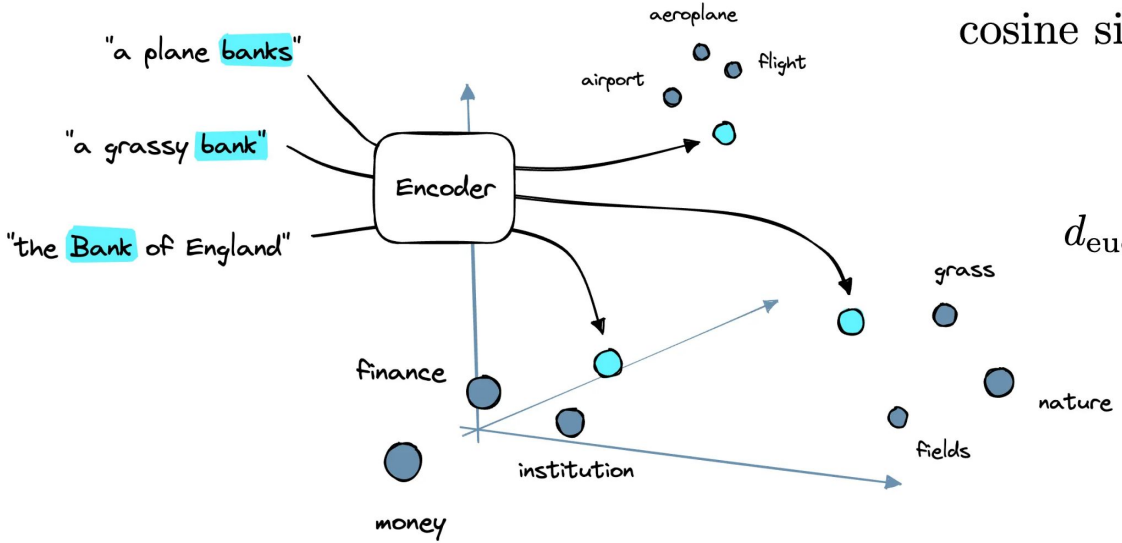








Text encoders



$$\text{cosine similarity}(x_1, x_2) = \frac{x_1 \cdot x_2}{\|x_1\| \, \|x_2\|}$$

$$d_{\text{euclid}}(x_1, x_2) = \sqrt{\sum_{i=1}^{n} (x_{1i} - x_{2i})^2}$$

Image from https://www.pinecone.io/learn/series/image-search/vision-transformers/

CLIP training

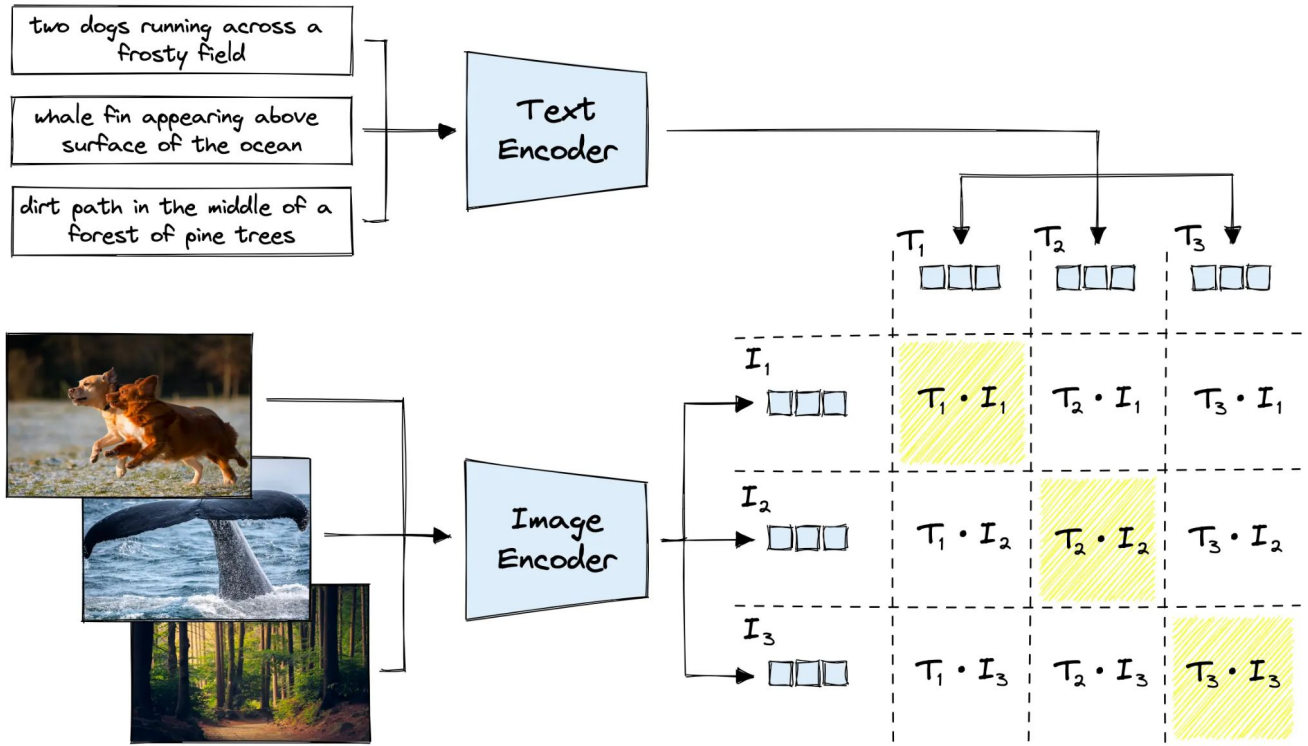


Image from https://www.pinecone.io/learn/series/image-search/clip/

NVIDIA’s C-RADIO for Text ↔ Image

Text Prompt: "shoe"

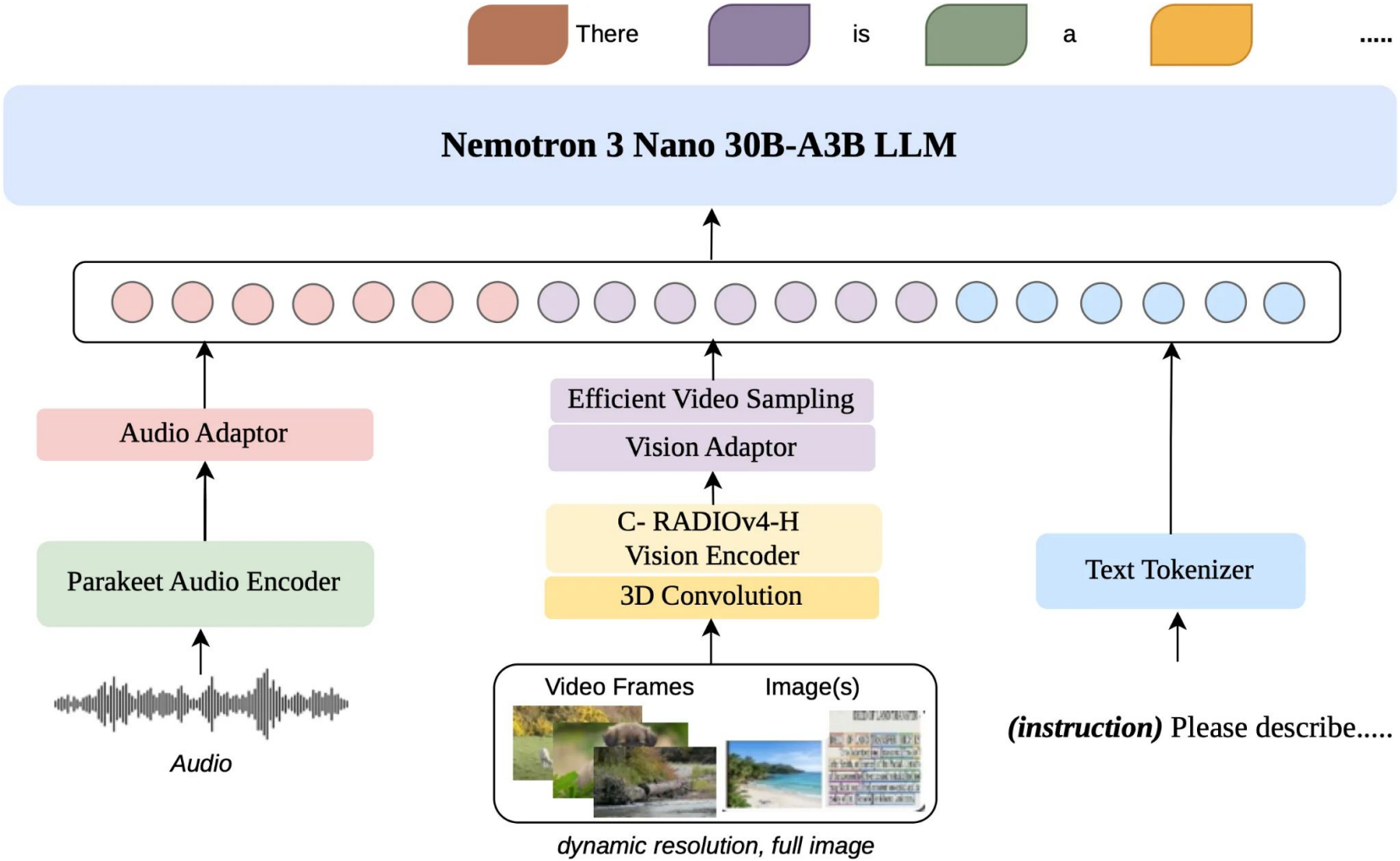


Text Prompt: "helmet"



https://huggingface.co/nvidia/C-RADIOv4-H

C-RADIOv4 (Tech Report)


There
is
a
.....
Nemotron 3 Nano 30B-A3B LLM
Audio Adaptor
Efficient Video Sampling
Vision Adaptor
C- RADIOv4-H
Vision Encoder
3D Convolution
Text Tokenizer
Parakeet Audio Encoder
Video Frames
Image(s)
Audio
dynamic resolution, full image
(instruction) Please describe.....

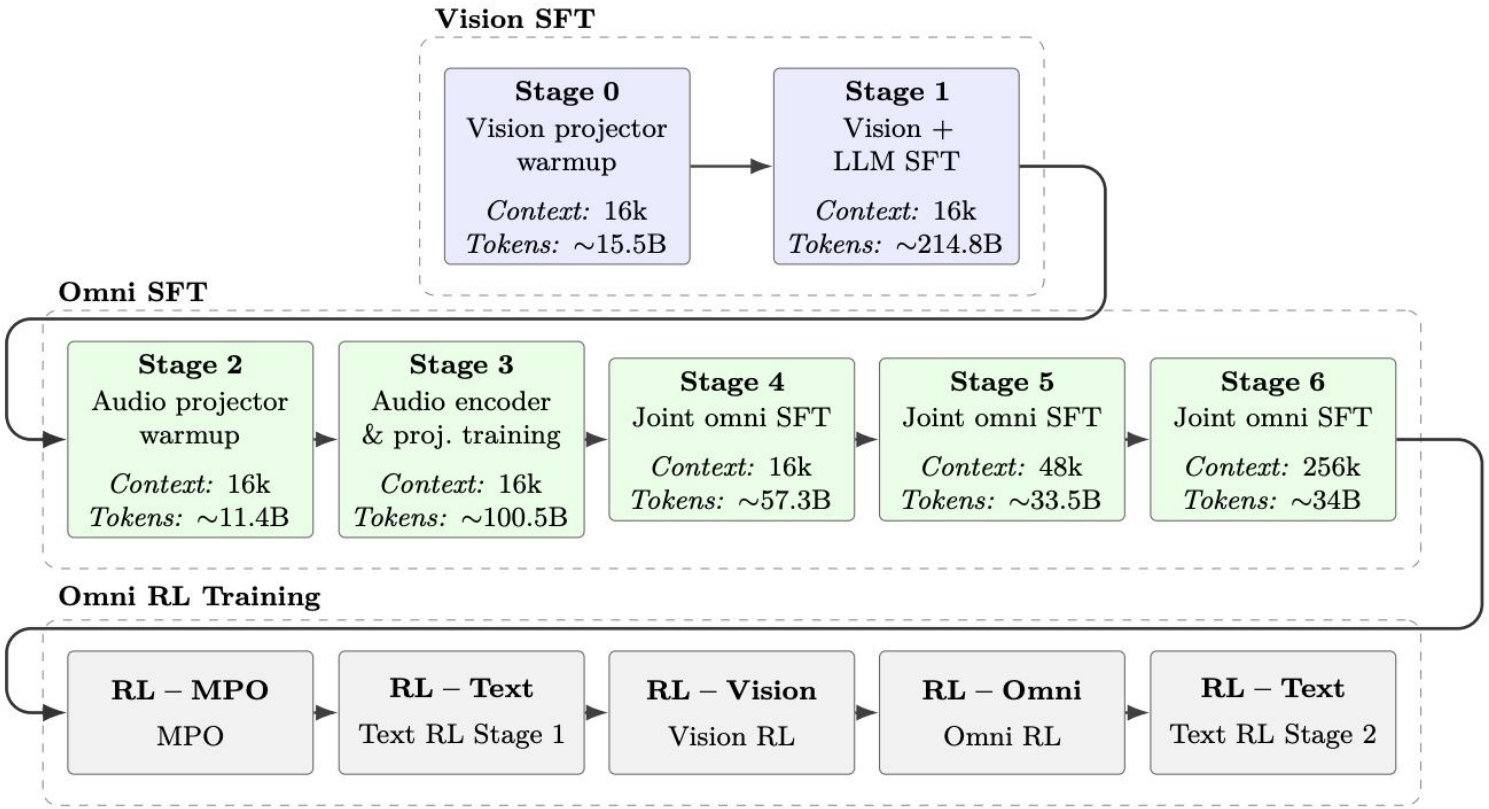


Figure 2 | Staged training recipe for the v3 omni-modal model. The pipeline first performs vision SFT, then joint omni SFT while progressively extending context length, followed by omni-modal RL training.

Full training recipe

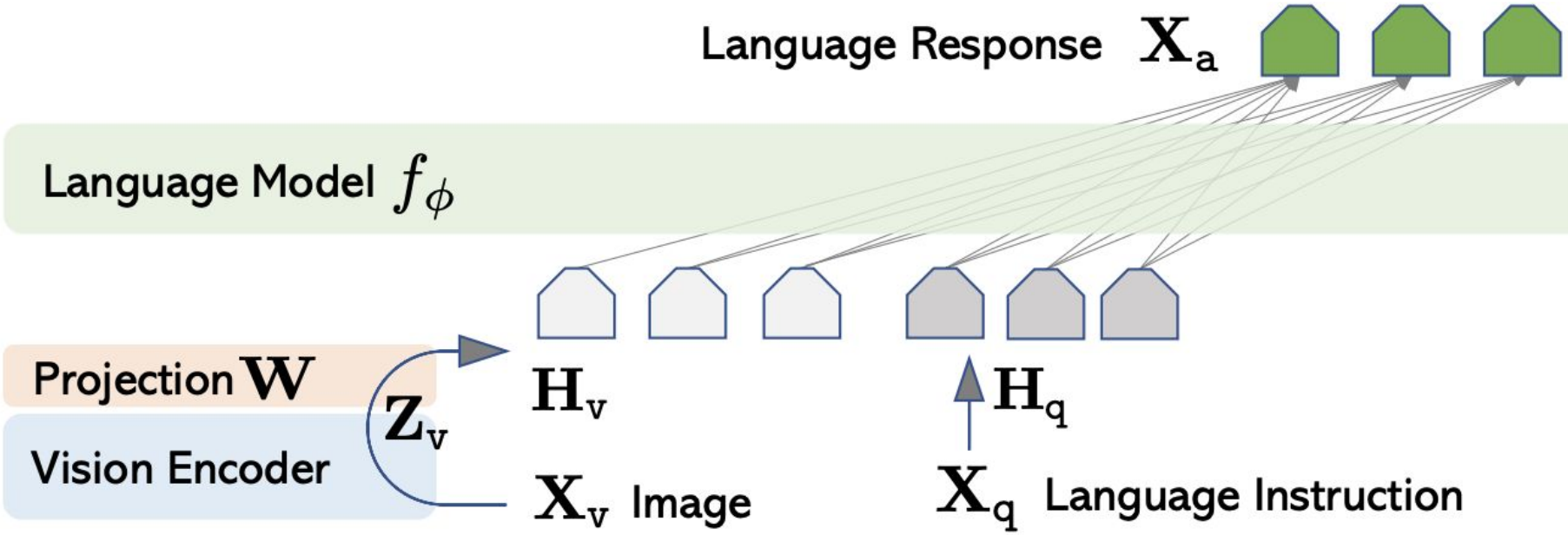


Figure 1: LLaVA network architecture.

Visual Instruction Tuning

Text Generation with Large Language Models

Auto-regressive prediction of tokens



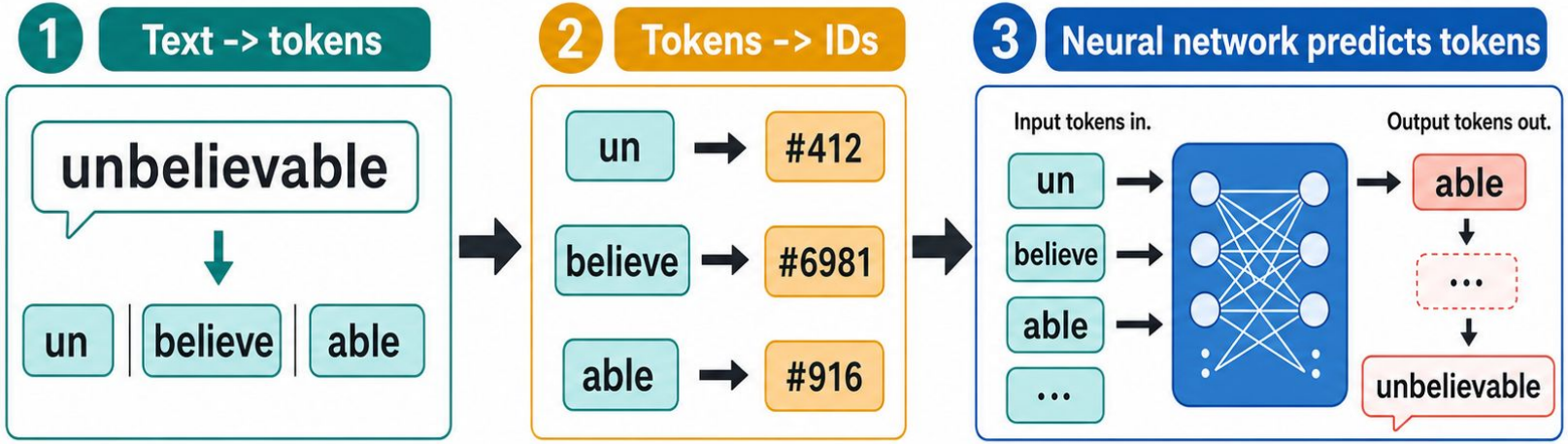
Deep Dive into LLMs like ChatGPT

What is a token?

A token is the unit of text (group of words, word, subword, or character) that is mapped to an integer ID and used as the input and output of the model.

Tokens are how LLMs read and write

Indexed subwords are both the input and output of the neural network.



Example 1: unbelievable -> un | believe | able

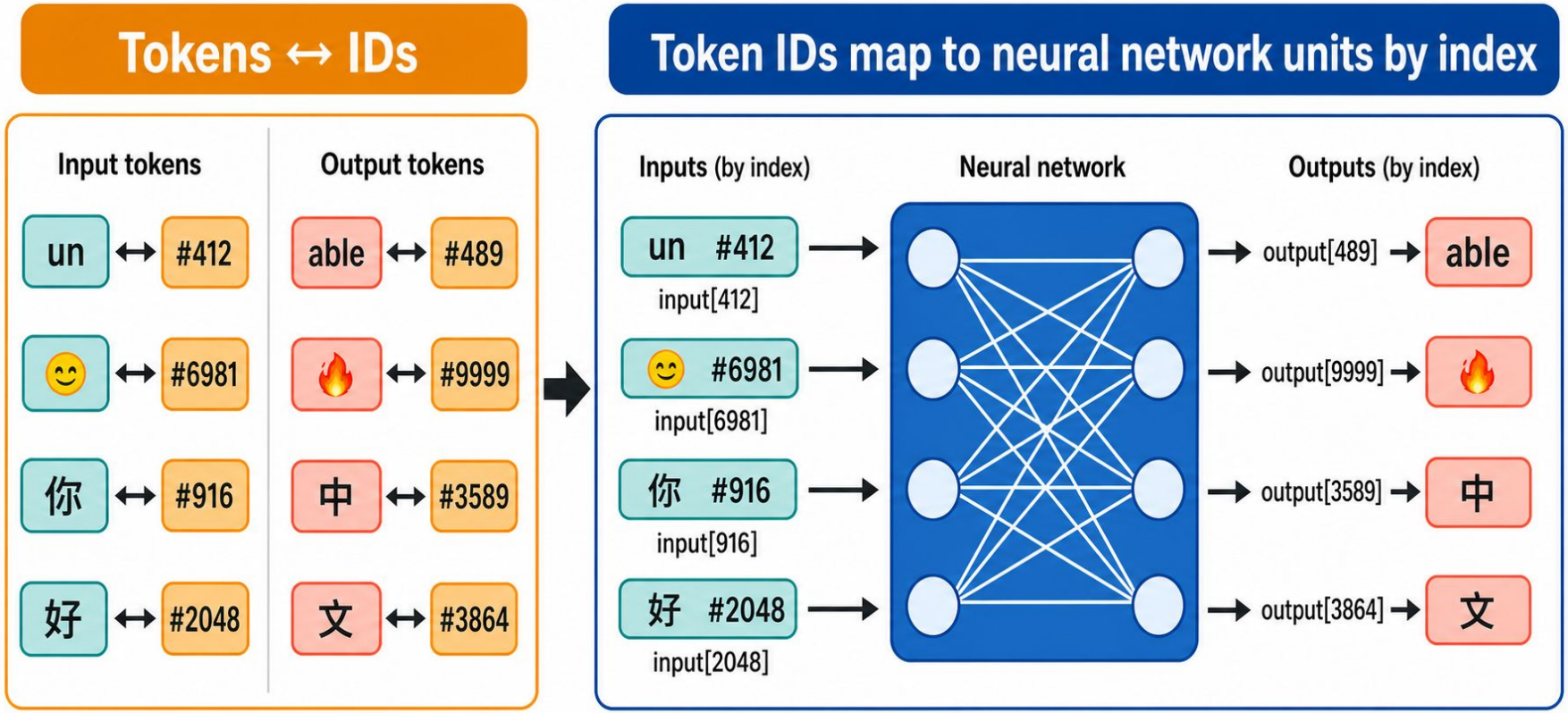
Example 2: un = #412 believe = #6981 able = #916

Caption: Input tokens in. Output tokens out.

LLMs do not process words directly; they process token indexes.

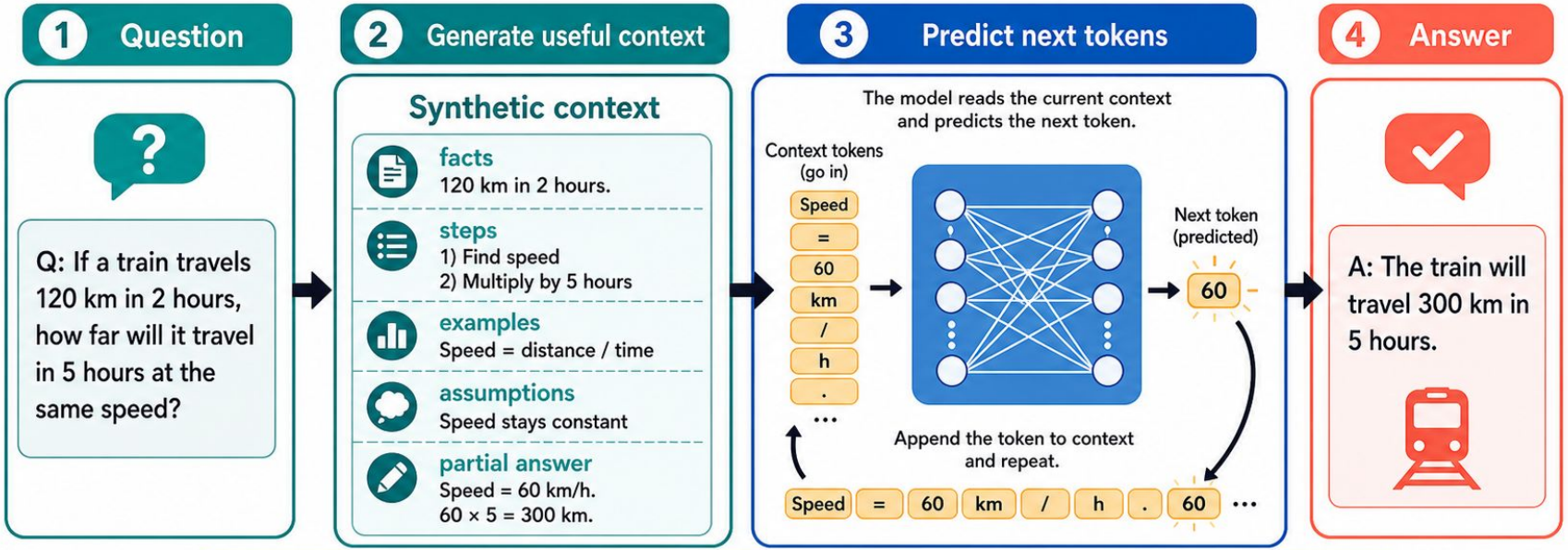
Let's reproduce GPT-2 (124M)

NVIDIA


Tokens ↔ IDs
Input tokens
Output tokens
un ↔ #412
able ↔ #489
😊 ↔ #6981
🔥 ↔ #9999
你 ↔ #916
中 ↔ #3589
好 ↔ #2048
文 ↔ #3864
Token IDs map to neural network units by index
Inputs (by index)
Neural network
Outputs (by index)
un #412
input[412]
😊 #6981
input[6981]
你 #916
input[916]
好 #2048
input[2048]
output[489] able
output[9999] 🔥
output[3589] 中
output[3864] 文

How do LLMs "reason"?

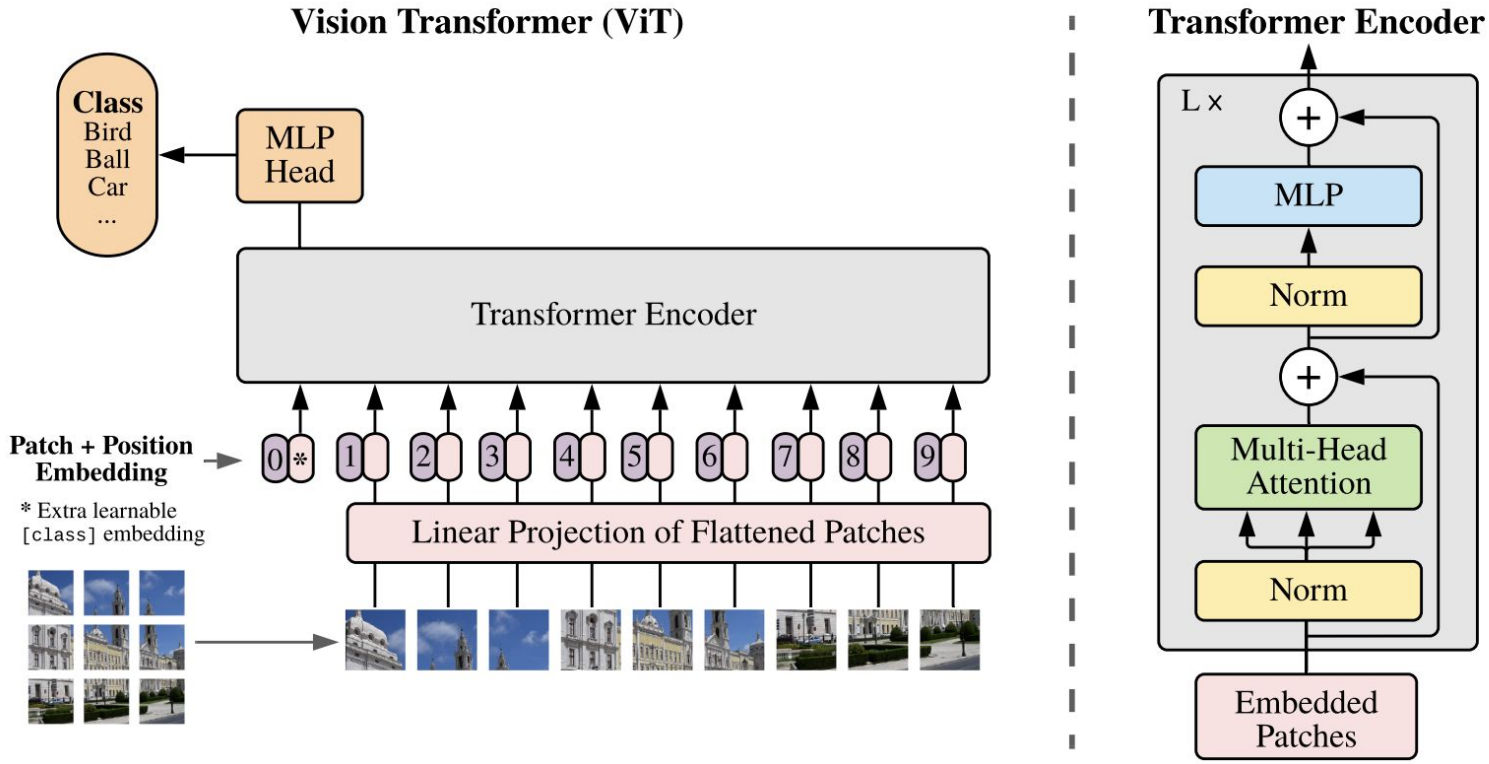
They produce synthetic context before answering.



Reasoning is more generated text: intermediate tokens that shape the final output.

The model is not pausing to think like a person; it is extending context token by token.

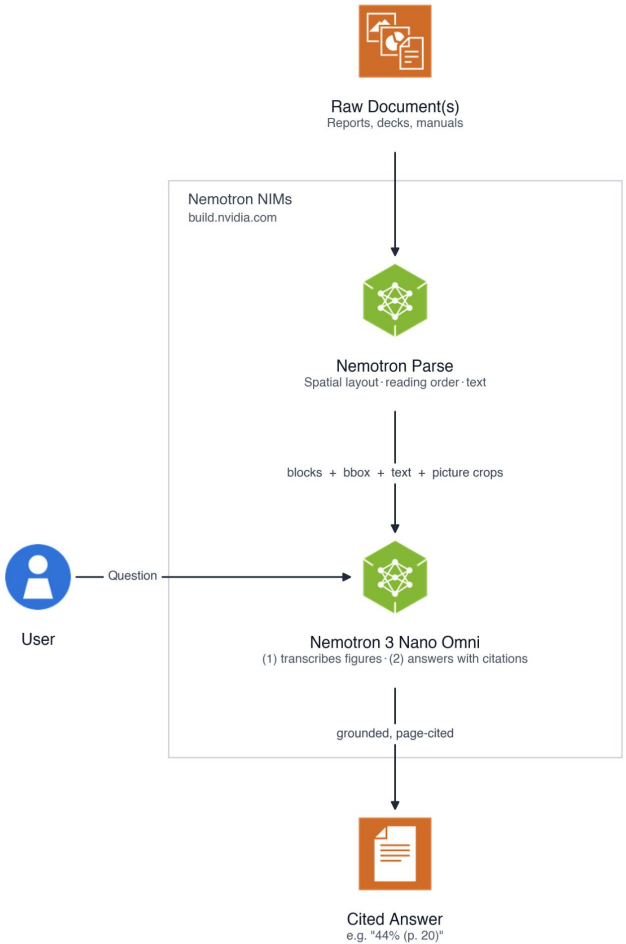
Chain-of-Thought Prompting Elicits Reasoning in Large Language Models



Vision Transformers



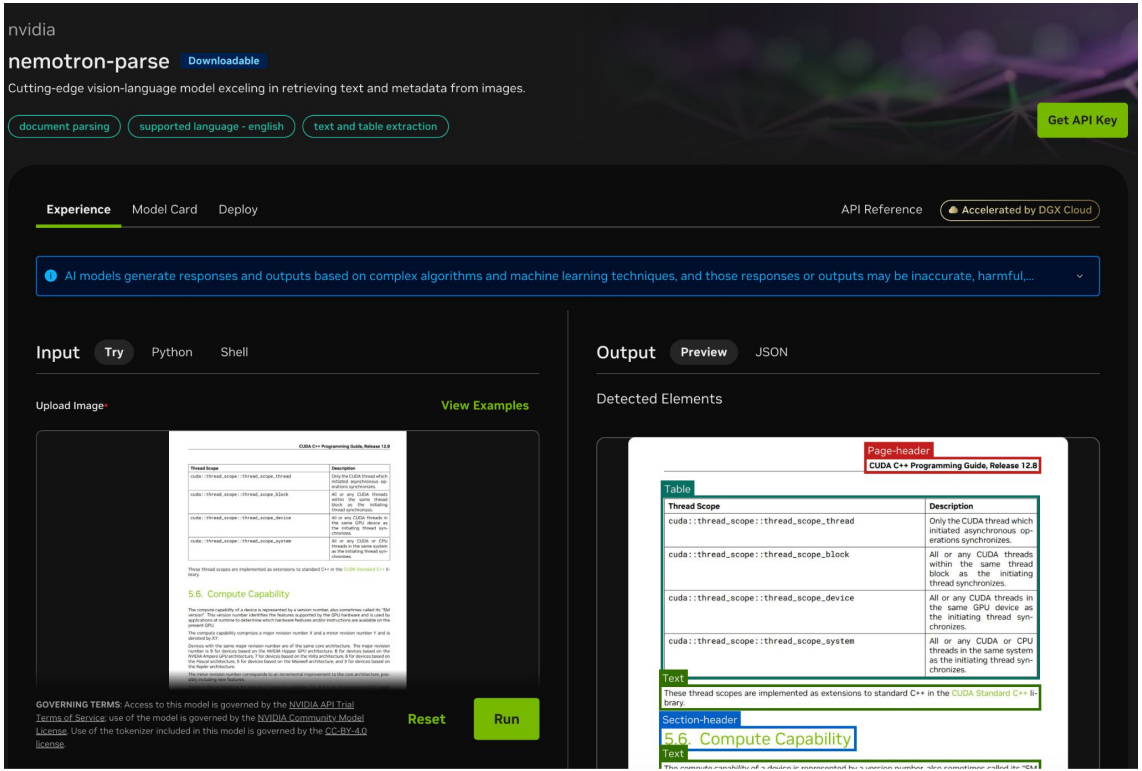
Would you call this “multi-modal”?



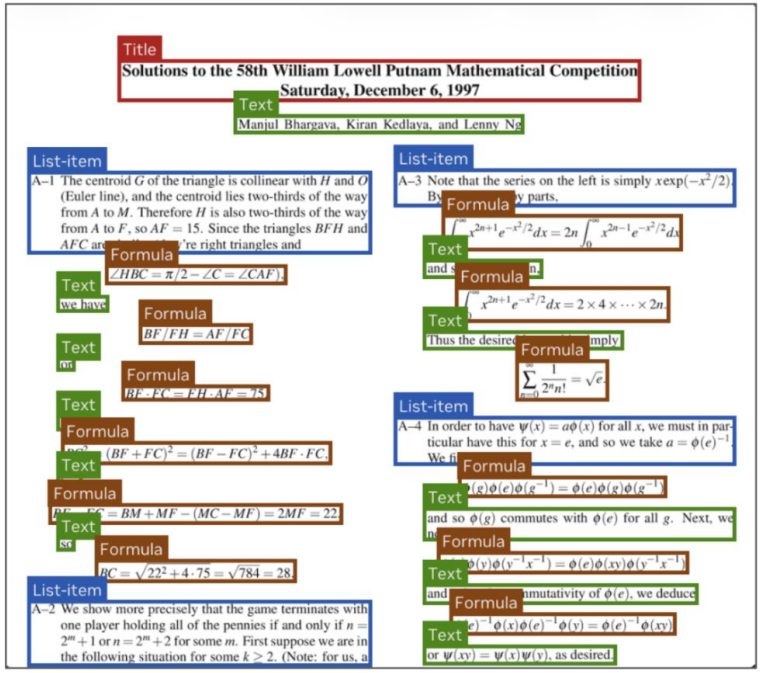
For complex OCR it’s best to pair with Nemotron Parse

Nemotron Parse

Creates additional context for Nemotron 3 Omni



https://build.nvidia.com/nvidia/nemotron-parse



Predicted semantic classes and bboxes

Solutions to the 58th William Lowell Putnam Mathematical Competition Saturday, December 6, 1997

Manjul Bhargava, Kiran Kedlaya, and Lenny Ng

A-1 The centroid G of the triangle is collinear with H and O (Euler line), and the centroid lies two-thirds of the way from A to M. Therefore H is also two-thirds of the way from A to F, so AF=15. Since the triangles BFH and AFC are similar (they're right triangles and

$\angle HBC = \pi/2 - \angle C = \angle CAF)$,

we have

$$BF/FH = AF/FC$$

or

$$BF \cdot FC = FH \cdot AF = 75.$$

Now

$$BC^2 = (BF + FC)^2 = (BF - FC)^2 + 4BF \cdot FC,$$

but

$$BF - FC = BM + MF - (MC - MF) = 2MF = 22,$$

so

$$BC = \sqrt{22^2 + 4 \cdot 75} = \sqrt{784} = 28.$$

A-2 We show more precisely that the game terminates with one player holding all of the pennies if and only if n=2^m+1 or n=2^m+2 for some m. First suppose we are in the following situation for some $k \geq 2$. (Note: for us, a

A-3 Note that the series on the left is simply $x\exp(-x^2/2)$. By integration by parts,

$$\int_0^\infty x^{2n+1} e^{-x^2/2} dx = 2n \int_0^\infty x^{2n-1} e^{-x^2/2} dx$$

and so by induction,

$$\int_0^\infty x^{2n+1} e^{-x^2/2} dx = 2 \times 4 \times \cdots \times 2n.$$

Thus the desired integral is simply

$$\sum_{n=0}^\infty \frac{1}{2^n n!} = \sqrt{e}.$$

A-4 In order to have $\psi(x) = a\phi(x)$ for all x, we must in particular have this for x=e, and so we take $a = \phi(e)^{-1}$. We first note that

$$\phi(g)\phi(e)\phi(g^{-1}) = \phi(e)\phi(g)\phi(g^{-1})$$

and so $\phi(g)$ commutes with $\phi(e)$ for all g. Next, we note that

$$\phi(x)\phi(y)\phi(y^{-1}x^{-1}) = \phi(e)\phi(xy)\phi(y^{-1}x^{-1})$$

and using the commutativity of $\phi(e)$, we deduce

$$\phi(e)^{-1}\phi(x)\phi(e)^{-1}\phi(y) = \phi(e)^{-1}\phi(xy)$$

or $\psi(xy) = \psi(x)\psi(y)$, as desired.

Rendered LaTeX Formulas and formatting

Figure 2 | OCR, extraction of text formatting and mathematical equations in LaTeX and markdown.

Nemotron Parse Technical Report

2.2.2. Output format

Nemotron-Parse predicts bounding boxes of the semantic blocks in the form of relative coordinates, in a scale of 1024×1280. These bounding boxes are predicted in a canonical reading order, that includes Page-Header elements at the start of the page, followed by Text, Section-Header, List-Item, Title and Formula elements in the order as they would be read by a person looking at the given page, followed by Footnotes, Page-Footers, Tables, Pictures, and Captions in the end. Nemotron-Parse-TC improves upon this canonical reading order, and also includes non-reading-order (floating) elements, i.e., Footnotes, Page-Footers, Tables, Pictures, and Captions within the natural ordering of the page.

In the maximal information setting the output is in the following format:

```
<x_(\d+)><y_(\d+)>(.*?)<x_(\d+)><y_(\d+)>
                        <class_([^>]+)>
```

where the the **first group** denotes the coordinates of the top-left corner, the **second group** denotes the text contained within the bounding box, the **third group** denotes the coordinates of the bottom-right corner, and the **final group** represents the semantic class. For example, `<x_0.1152><y_0.2586># NVIDIA Nemotron-Parse 1.1<x_0.8799><y_0.2797> <class_Title>`.

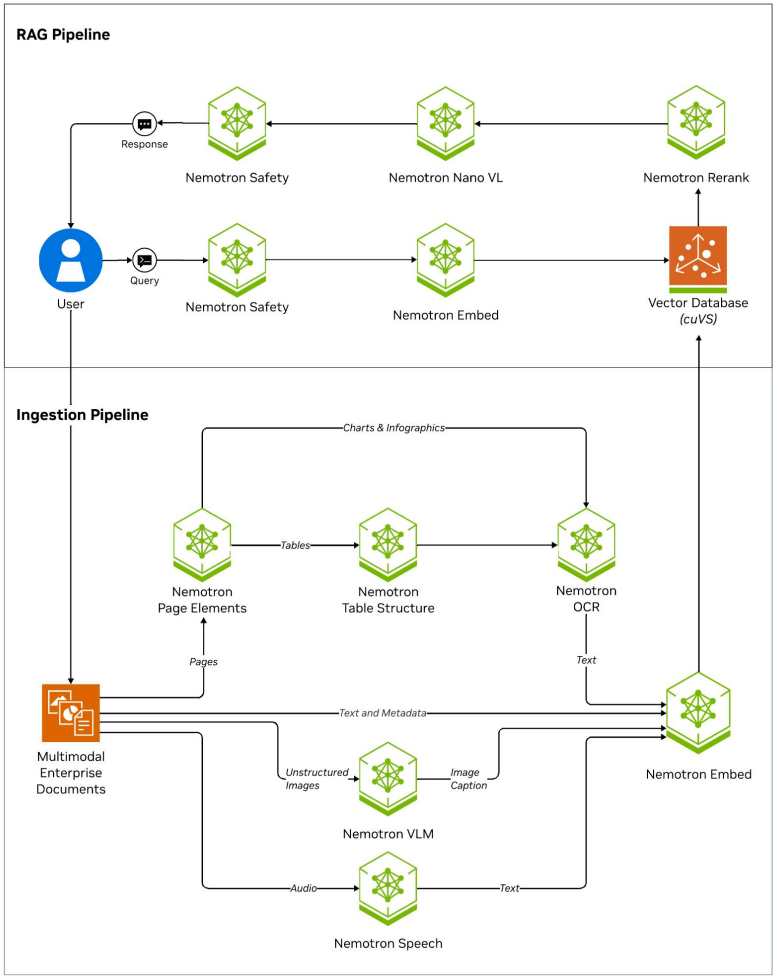
4.5. Reasoning budget control

We study the effect of inference-time reasoning budgets by evaluating model performance under two settings: (1) a base configuration with a maximum sequence length of 16,384 tokens, and (2) a reasoning-enabled configuration with a 13K reasoning budget, a 1,024-token grace period, and a maximum sequence length of 16,384 tokens (Table 11).

Our results suggest that reasoning budget adjustment yields accuracy gains on select benchmarks under reasoning-on mode, with no degradation observed on the remaining ones. These gains with budget control may arise from the early termination of malformed reasoning traces with repetition loops on out-of-distribution tasks, as well as the truncation of overly verbose reasoning chains for problems requiring minimal or straightforward reasoning.

Benchmark	MathVista-Mini	MMLongBench-Doc	DocVQA (Val)	Charxiv(RQ)	RefCOCO	VideoMME
w.o. reasoning budget	80.3	54.5	**95.3**	61.8	90.4	67.5
w. reasoning budget	**82.8**	**56.8**	95.2	**64**	**90.6**	**70.3**

Table 11 | Effect of reasoning budget across several key benchmarks.



[Nemo Retriever]


OK, SO WHAT IS 'BEST'?
BOOKS [0.94]
COMPUTER [0.96]
SEARCH RESULTS [0.92]
ARTICLE [0.89]
TEXT [0.95]
GUY [0.98]
WINDOW [0.82]
NEWSPAPERS [0.89]
DIAGRAMS [0.91]
DESK [0.95]
MAPS

“It depends…”

- Vidore v3
- Parse Bench
- AEC Bench

Summary

Nemotron 3 Omni unifies text, vision, and audio in a single model

- Embeddings enable flexible multimodal workflows
- Two modes: descriptive and reasoning

Powers end-to-end pipelines for document intelligence, multimodal search, and assistants

- Multimodal prompts let the model attend to specific regions, layouts, or visual details
- Integrates with tools like Nemotron Parse and Nemo Retriever to build production-ready multimodal apps
- No Free Lunch, you have to try it on your data but Vidore v3, Parse Bench, and AEC Bench are places to start looking

Resources

- Nemotron 3 Nano Omni - Technical Report
- Nemotron Developer repository at GitHub
- Nemotron 3 Nano Omni 30B @ HuggingFace
- Nemotron Parse 2.0 at build.nvidia.com
- Fine-tuning recipe for Nemotron Parse
- Vidore v3 Benchmark
- Practical Computer Vision Bootcamp - GitHub Repository

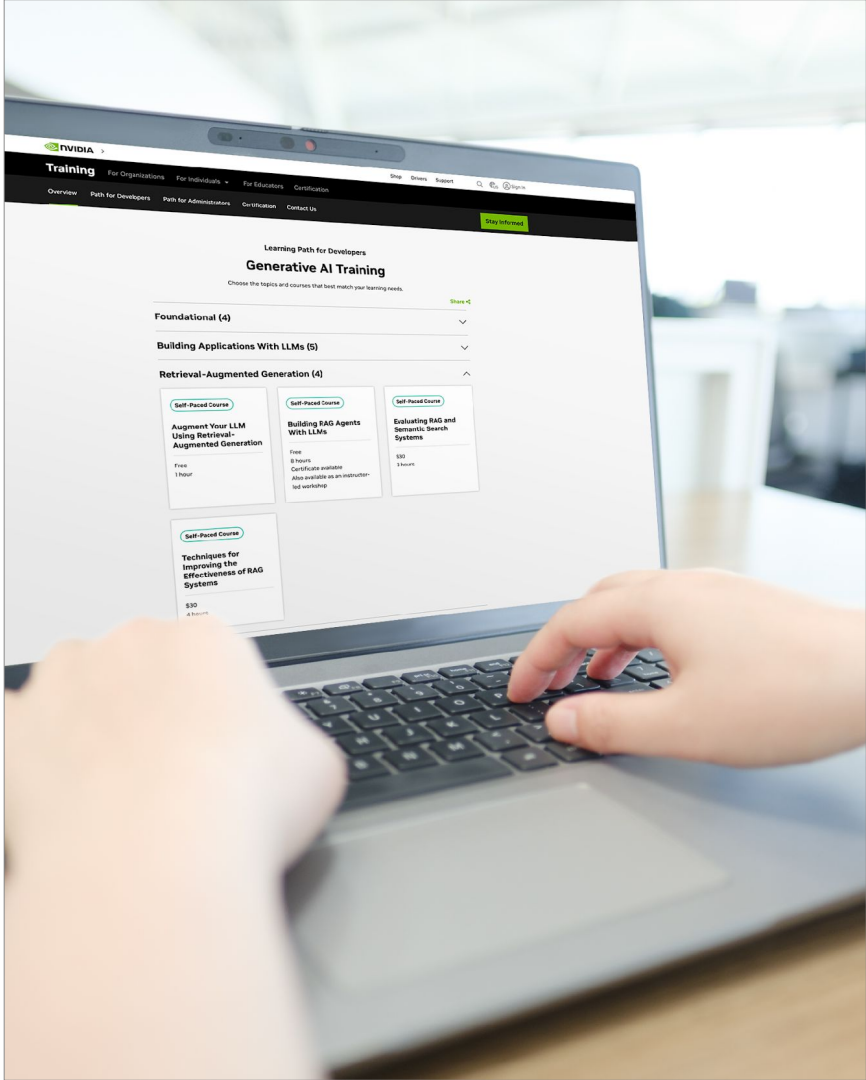
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